

Assignment 04

Part 01:

1. Considering the Code Below, Write Down the Body of all Listed Methods and Properties and Constructor:

```
public class Book
{
    public string ISBN { get; set; }
    public string Title { get; set; }
    public string[] Authors { get; set; }
    public DateTime PublicationDate { get; set; }
    public decimal Price { get; set; }

    public Book(string _ISBN , string _Title ,
               string[] _Authors , DateTime _PublicationDate ,
                                   decimal _Price)
    {
        throw new NotImplementedException();
    }

    public override string ToString()
    {
        throw new NotImplementedException();
    }
}
```

```

public class BookFunctions
{
    public static string GetTitle (Book B)
    {
        throw new NotImplementedException();
    }

    public static string GetAuthors(Book B)
    {
        throw new NotImplementedException();
    }

    public static string GetPrice(Book B)
    {
        throw new NotImplementedException();
    }
}

```

1. You need to parameterize **ProcessBooks** function to accept **BookFunctions** Methods using following cases:

- Create User Defined Delegate with the same signature of methods existed in Bookfunctions class.
- Use the Proper build in delegate.
- Anonymous Method (GetISBN).
- Lambda Expression (GetPublicationDate).

```

public class LibraryEngine
{
    public static void ProcessBooks ( List<Book> bList
                                   ,/*Pointer To BookFunciton*/ fPtr)
    {
        foreach (Book B in bList)
        {
            Console.WriteLine(fPtr(B));
        }
    }
}

```

Part 02:

2. **We need to Implement the List methods from scratch with all overloads.**

- Exist
- Find
- Find All
- Find index
- Find Last
- Find Last Index
- Foreach
- TrueForAll